

ASHNA NAWAR AHMED

Education

Texas State University, San Marcos, Texas, USA

Ph. D. in Computer Science

Fall 2024 (Aug 2024) – present

ONGOING

Ahsanullah University of Science and Technology, Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering

Fall 2016 – Spring 2020

Nov 2016 – Aug 2021

CGPA: 3.871/4.00 [1st in Merit Position]

Adamjee Cantonment College, Dhaka, Bangladesh

Higher Secondary Certificate (HSC) in Science Group

Jul 2014 – Oct 2016

GPA: 5.00/5.00

Mohammadpur Preparatory Higher Secondary School, Dhaka, Bangladesh

Secondary School Certificate (SSC) in Science Group

Jun 2014

GPA: 5.00/5.00

Research Interests

AI/ML for HPC Systems, Multi-Objective Optimization, Surrogate Modeling, Job Scheduling, Energy-Aware Computing, High Performance Computing, Neural Networks, Evolutionary Algorithms

Member of the Per4ML Research Group (Supervisor: Dr. Tanzima Islam, Associate Professor, Texas State University) — Pioneering work on intelligent, data-driven optimization for sustainable HPC scheduling.

Publication

Attention-Informed Surrogates for Navigating Power-Performance Trade-offs in HPC

Machine Learning for Systems (MLForSys) Workshop, NeurIPS 2025 | Accepted: September 2025

- Accepted workshop paper at NeurIPS 2025, proposing attention-informed surrogate models for efficient multi-objective HPC optimization.

Intelligent Surrogates Pay Attention to Data, Improving Multi-Objective HPC Optimization

Link | Accepted: September 2025

- Accepted poster at SC25, showcasing data-driven surrogate models for energy-aware HPC scheduling.

Decentralizing and Optimizing Nation-Wide Employee Allocation While Simultaneously Maximizing Employee Satisfaction

DOI link | Published: Dec 2022

- The goal was to decentralize the workers of Bangladesh and allocate them evenly in every district of Bangladesh where their satisfaction will be maximized while ensuring maximum dispersion using Multi-Objective Optimization and Neural Network.

Experience

Oak Ridge National Laboratory

Intern, Graduate Research at ORNL (GRO) program, Supervisor: Terry Jones, Computer Scientist Knoxville, Tennessee, USA

July 2025 – August 2025

Texas State University

Doctoral Research Assistant, Department of Computer Science, Supervisor: Dr. Tanzima Islam

Aug 2024 – Present

San Marcos, Texas, USA

Ahsanullah University of Science and Technology

Lecturer (On Study Leave), Department of Computer Science & Engineering

Nov 2021 – July 2024

Dhaka, Bangladesh

Ahsanullah University of Science and Technology

Part-time Lecturer, Department of Computer Science & Engineering

Sep 2021 – Oct 2021

Dhaka, Bangladesh

Certifications

International English Language Testing System (IELTS) Overall Band: **8.5**

Speaking **8.5**, Reading **9.0**, Listening **8.5**, Writing **7.5** | Result published on: August 1, 2023

Graduate Record Examinations (GRE) Overall Score: **325**

Quantitative Reasoning: **161**, Verbal Reasoning **164**, Analytical Writing **4.0** | Test Taken on: October 11, 2023

Training For Mobile Application Developer (iOS): Skill Development for Mobile Game and Application Project

Conducted by - ICT Division, Bangladesh | *May 2018 - Nov 2018*

Academic Projects

Information Pool System Website | *C#; Framework: ASP.NET MVC* [Git](#) **2019**

Open Credit System: Bootstrap Website | *HTML, PHP, Javascript* [Git](#) **2019**

Open Credit System: Database Project | *Java, SQL Server* [Git](#) **2018**

Memory Lane App(notes, recordings, alarms) | *Android Studio, C++* [Git](#) **2018**

Matchmaking Desktop Software | *Java, Swing, Netbeans* [Git](#) **2017**

Colour Switch Game | *iGraphics Framework, C, Visual Studio* [Git](#) **2017**

Awards and Achievements

Invitation to join Phi Kappa Phi Honor Society for academic excellence (2025).

1st Merit Position Holder (based on B.Sc. result, in a batch of 75 people)

Dean's List of Honor (based on B.Sc. result): Spring 2020

1st Runners-Up: Intra-AUST Project Showcase Spring 2018 (LFR)

Top 40 in National Girls' Programming Contest 2018

Technical Skills

Languages: Python, C, C++, C#, Java, HTML5, PHP, JavaScript, Swift

Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, BoTorch, DeepSpeed, FairScale, TabNet, MPI, OpenMP, CUDA, ASP.NET MVC, Bootstrap, WebGL, CSS

Databases: MySQL, MS SQL Server, SQLite, Oracle

Design & Development Tools: Eclipse, NetBeans, Visual Studio, PyCharm, Jupyter Notebook, Xcode

Operating Systems: Linux, Windows, Android, iOS

Others: HPC job schedulers (Slurm, TACC environment), Git, LaTeX, PL/SQL

References

[Dr. Tanzima Islam](#)

Assistant Professor

Department of Computer Science

Texas State University

✉ tanzima@txstate.edu

[Terry Jones](#)

Senior Research Staff (Computer Scientist)

Computer Science and Mathematics Division (CSMD)

Oak Ridge National Laboratory (ORNL)

✉ trjones@ornl.gov